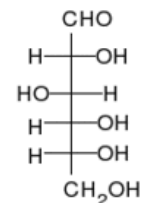
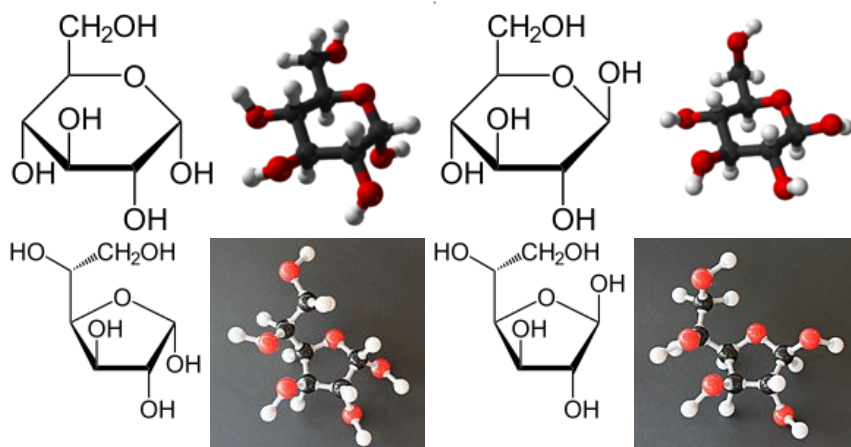


arrangement), while " β -" means that they are on the same side of the plane (a *cis* arrangement). Therefore, the open-chain isomer D-glucose gives rise to four distinct cyclic isomers: α -D-glucopyranose, β -D-glucopyranose, α -D-glucofuranose, and β -D-glucofuranose. These five structures exist in equilibrium and interconvert, and the interconversion is much more rapid with acid catalysis.

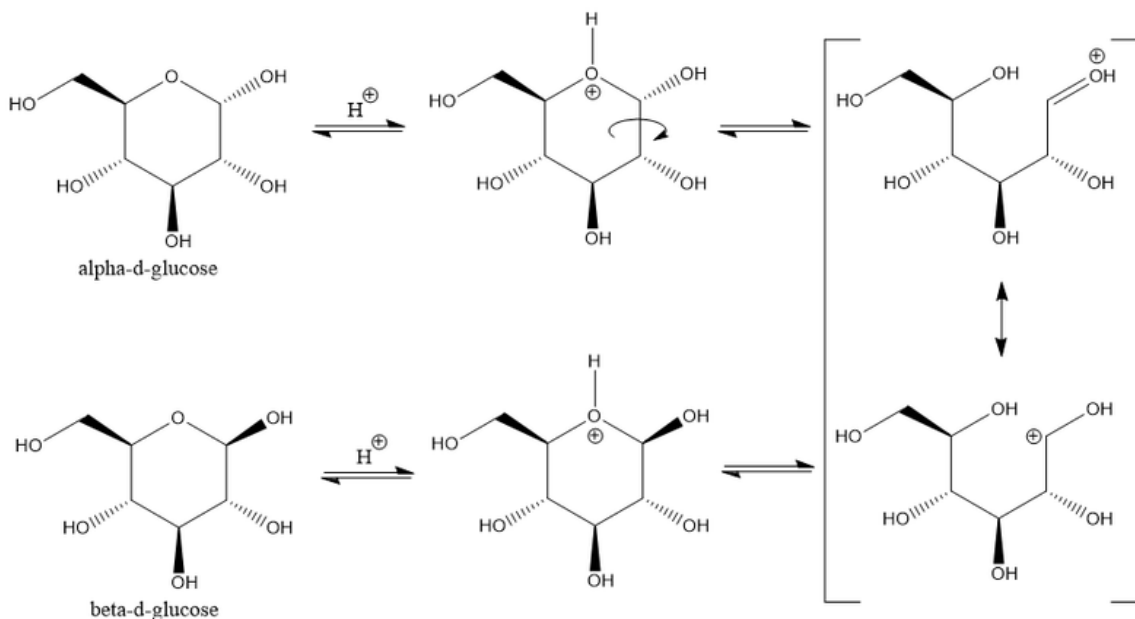


Glucose can exist in both a straight-chain and ring form.

Cyclic forms of glucose



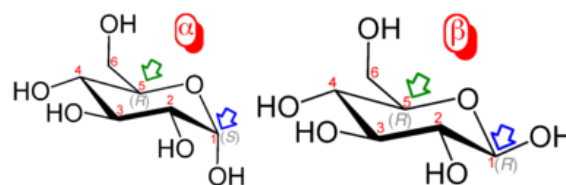
From left to right: Haworth projections and ball-and-stick structures of the α - and β - anomers of D-glucopyranose (top row) and D-glucofuranose (bottom row)



The other open-chain isomer L-glucose similarly gives rise to four distinct cyclic forms of L-glucose, each the mirror image of the corresponding D-glucose.

The rings are not planar, but are twisted in three dimensions. The glucopyranose ring (α or β) can assume several non-planar shapes, analogous to the "chair" and "boat" conformations of cyclohexane. Similarly, the glucofuranose ring may assume several shapes, analogous to the "envelope" conformations of cyclopentane.

In the solid state, only the glucopyranose forms are observed, forming colorless crystalline solids that are highly soluble in water and acetic acid but poorly soluble in methanol and ethanol. They melt at 146 °C (295 °F) (α) and 150 °C (302 °F) (β), and decompose at higher temperatures into



Chair conformations of α - (left) and β - (right) D-glucopyranose